

Representative Output Samples Illustrating the Performance Spectrum of LLMs

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1 Output Samples Illustrating the Performance Spectrum of LLMs

The following example outputs were deliberately selected, based on the seven categories in Table 2 of the article, to illustrate both the strengths and limitations of LLMs in solving ordinary differential equations. Observed results may vary depending on the specific LLM model employed.

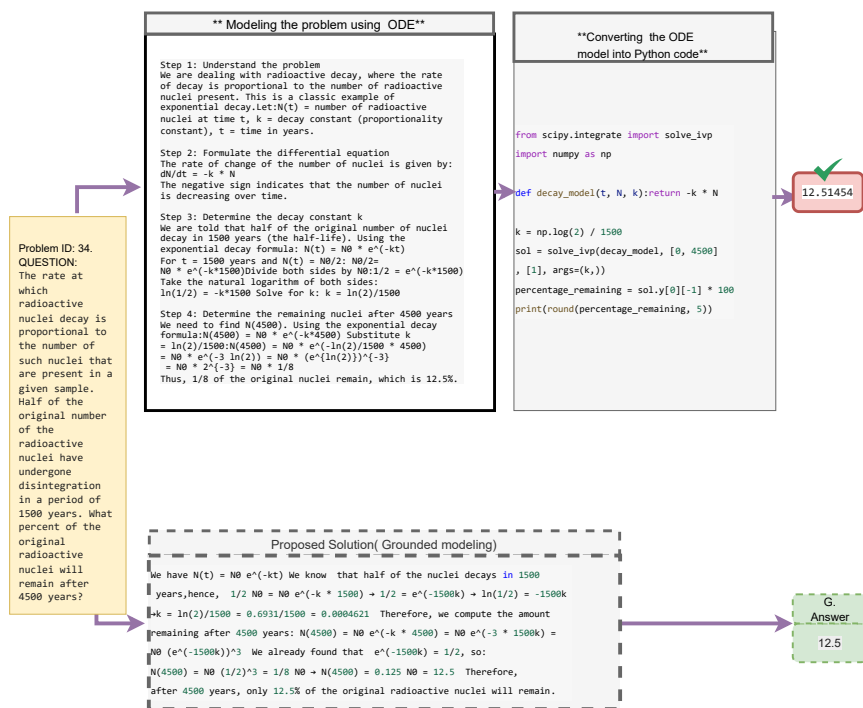


Figure 1: Example 1-Category:Chemical Kinetics Problem

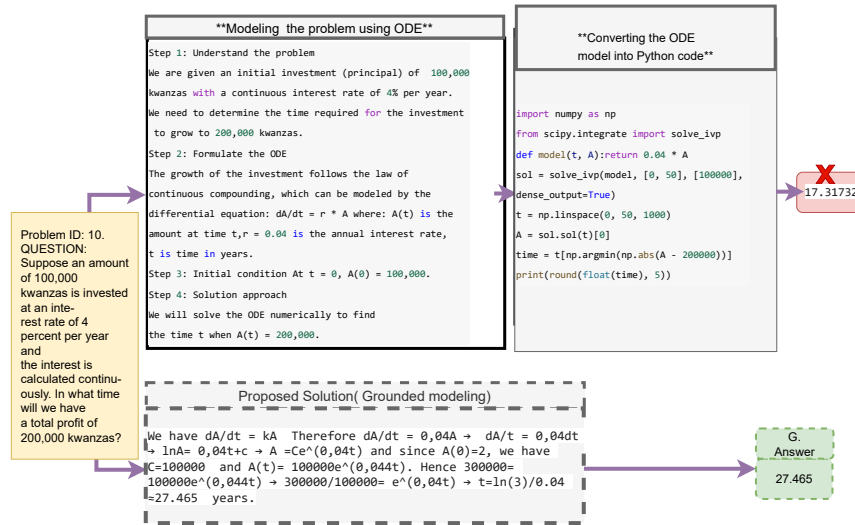


Figure 2: Example 2-Category: Economy Problem

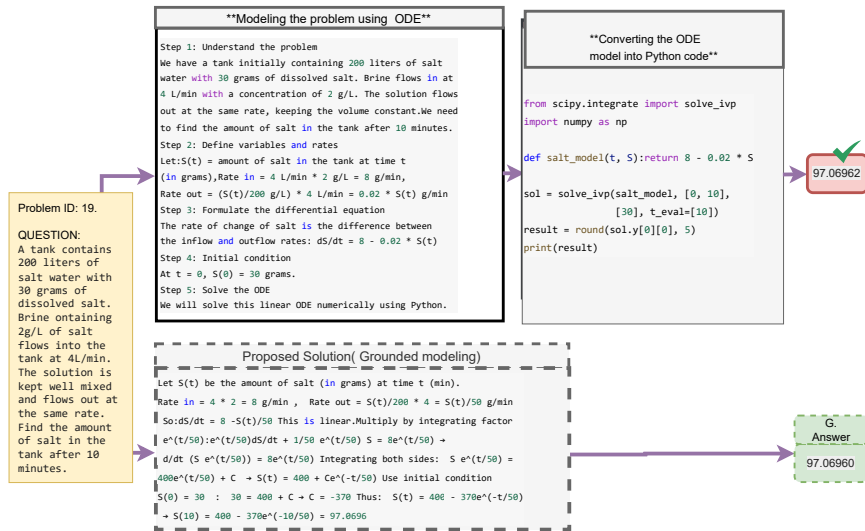


Figure 3: Example 3-Category: Mixing Problem

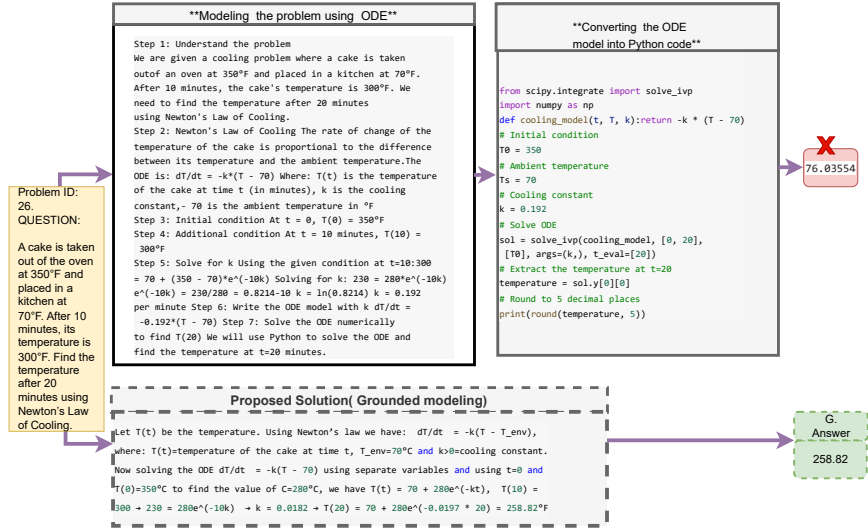


Figure 4: Example 4-Category: Newton's Law of Cooling Problem

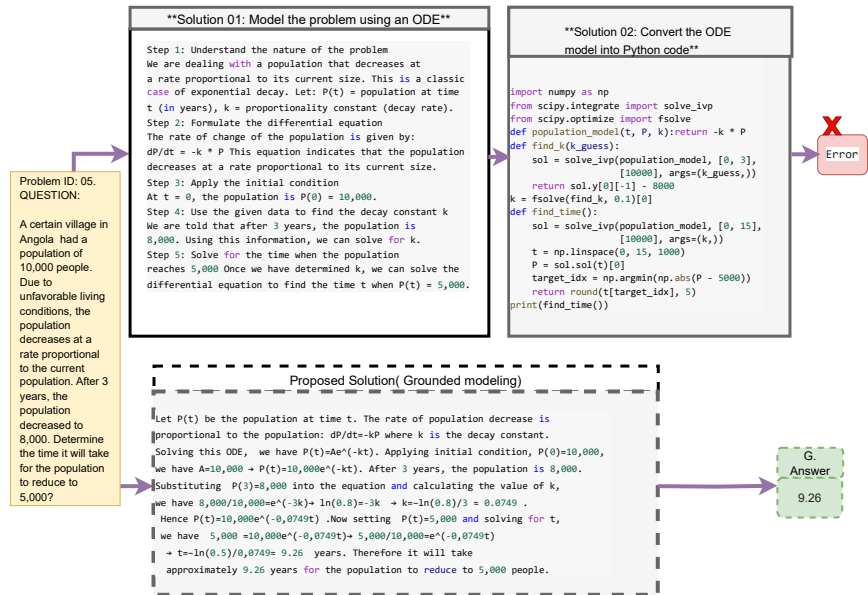


Figure 5: Example 5-Category: Exponential Growth/Decay Problem

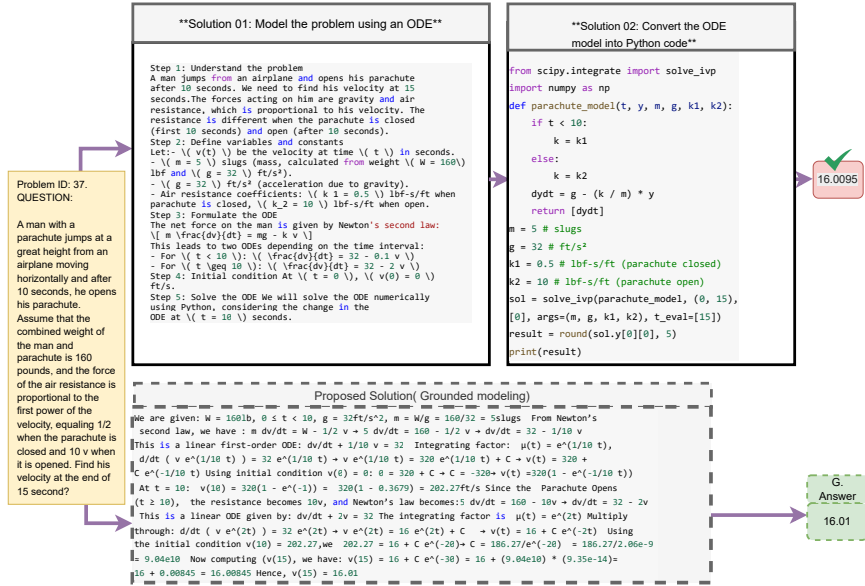


Figure 6: Example 6-Category: Falling Objects with Air Resistance Problem

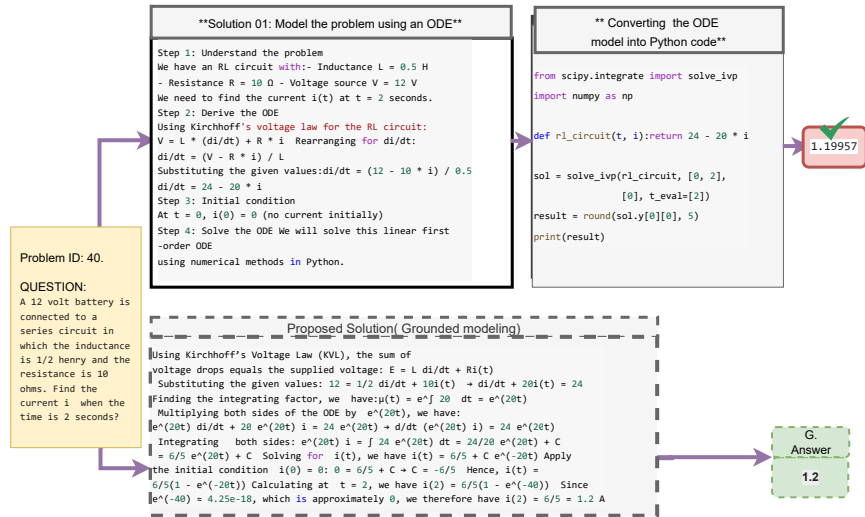


Figure 7: Example 7-Category: Electric Circuits Problem